MIL-G-17594B(SHIPS)
16 October 1967

SUPERSEDING
MIL-G-17594A(SHIPS)
8 February 1963
(See 6.3)

MILITARY SPECIFICATION

GROMMET, PLASTIC, UNICELLULAR, HIGH DENSITY

1. SCOPE

1.1 This specification covers one type of buoyant, high density, foamed plastic grommet of the shape and dimensions specified on Drawing S8225(5)-1170434.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
C-C-91 - Candles, Illuminating.

MILITARY
MIL-P-116 - Preservation, Method of
MIL-C-17415 - Cloth, Coated, and Webbing, Inflatable Boat and Miscellaneous Use.

STANDARDS

FEDERAL
FED-STD-595 - Colors.
FED-STD-601 - Rubber: Sampling and Testing.

MILITARY
MIL-STD-293 - Visual Inspection Guide for Cellular Rubber Items.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

DRAWING

NAVAL SHIP ENGINEERING CENTER
S8225(5)-1170434 - Inflatable Boat Equipment-Rescue Line-Details and Arrangements.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
D-471 - Change in Properties of Elastomeric Vulcanizates Resulting from Immersion in Liquids.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical Society and Technical Association Specifications and Standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Plastic. The plastic material shall be a poly (vinyl chloride) homo or copolymer containing phosphate plasticizers and compounded and pigmented to meet all the applicable requirements of this specification. Evidence that the resin and plasticizer used in formulating the plastic meets this requirement shall be provided as specified in 4.3.1.1.

3.1.2 Form. The plastic material shall be in the form of a flexible foam, composed of minute, evenly distributed closed cells of uniform size.

3.2 Finished grommet.

3.2.1 Physical properties. The finished grommet shall conform to the properties of table I.

Table I - Physical properties.

Property	Requirement		Test Reference
	Minimum	Maximum	
Density, pounds per cubic foot	25	55	4.4.1
Compression resistance at 25 percent deflection, pounds	40	80	4.4.2
Buoyancy net, pounds	0.1	--	4.4.3
Weight, grams	160	190	4.4.4
Strength in tension, initial pounds	200	--	4.4.5
Strength in tension, after 1 week at 165°F., pounds	150	--	4.4.6
Volume loss on heat aging after 1 week at 165°F., percent	---	5	4.4.7
Blocking after 1 week at 165°F.	No Blocking		4.4.8
Oil resistance	No Swelling		4.4.9
Fire resistance - flame duration, seconds	---	30	4.4.10
Fire resistance - length of char, inches	---	3	4.4.10

3.2.2 Color. The color shall be orange, between textile sea rescue color and color #12197 of FED-STD-595. The grommet shall be uniformly pigmented throughout.

3.2.3 Shape and dimensions. The plastic grommet shall conform to the shape and dimensions specified in Drawing S822S(S)-1170434.

3.2.4 Construction. The ring forming the grommet shall be molded in one piece and shall have a natural skin. Finishing operations are restricted to trimming the mold flash.

3.2.5 Odor. The grommet shall be free of objectionable odor.

3.3 Workmanship. The grommets shall be of uniform workmanship and shall be free from imperfections which may affect their serviceability or appearance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Sampling.

4.2.1 Lot. For purposes of sampling, examination and tests a lot shall consist of all grommets of the same size and composition, not exceeding 1,000, offered for delivery at one time.

4.2.2 Sampling for visual and dimensional examination. A random sample of grommets shall be selected from each lot in accordance with the sampling plan of table II for the examination specified in 4.3.1.

Table II - Sampling for visual and dimensional examination
AQL (approx.) = 1.5 percent defective.

Number of grommets in an inspection lot	Number of grommets in sample	Acceptance number (defectives)	Rejection number (defectives)
15 and under	10	0	1
16 to 40	15	0	1
41 to 110	25	1	2
111 to 300	35	1	2
301 to 500	50	2	3
501 to 800	75	3	4
801 to 1000	110	4	5

4.2.3 Sampling for tests. A random sample of grommets shall be selected in accordance with table III from each lot that passes the examination of 4.3.1. Quality conformance tests shall be conducted as specified in 4.3.2.

Table III - Sampling for tests.

Number of grommets in inspection lot	Number of grommets to be tested (except blocking 4.4.8)	4.4.8 (Blocking)
		(Both sample units must pass)
up to 8	None	2
9 to 15	2 (both must pass)	2
16 to 25	3	2
26 to 50	4	2
51 to 90	5	2
91 to 150	7	2
151 to 500	10	2
501 to 1000	15	2

4.3 Quality conformance inspection.

4.3.1 Visual and dimensional examination. Each of the sample grommets selected in accordance with table II shall be visually and dimensionally examined to verify compliance with the requirements for color (3.2.2), shape and dimensions (3.2.3), construction (3.2.4) odor (3.2.5), and workmanship (3.3). Each of the sample grommets shall also be examined for the defects listed in MIL-STD-293, as applicable. Any grommet in the sample containing one or more visual or dimensional defects shall not be offered for delivery, and if the number of defective grommets in the sample exceeds the acceptance number of table II, the lot represented by the sample shall be rejected.

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4.3.1.1 Certification of composition. Contractor shall furnish certification that the resin used in formulating the plastic material (3.1) is a homo or copolymer or poly (vinyl chloride) and that phosphate plasticizers are used.

4.3.2 Quality conformance tests. Quality conformance tests shall be conducted on samples from each lot offered for delivery in accordance with table III. All the tests specified in 4.4 shall be performed. The data obtained from each test, except 4.4.8, shall be used as shown in MIL-STD-414.

4.3.3 Action in case of nonconformance. If the number of defectives in the samples examined under 4.3.1 or in the quality conformance tests (4.3.2) exceeds the acceptable number, the lot which it represents shall be rejected. The contractor may resubmit the rejected lot only after removal of all defective grommets. Resubmitted lots shall be sampled and tested as specified in 4.2 and 4.3.2.

4.4 Test procedures.

4.4.1 Density. The density of the grommet shall be determined by dividing the weight of the grommet by its volume and shall be expressed in pounds per cubic foot. The volume shall be determined by measuring the volume of water displaced by the grommet.

4.4.2 Compression resistance at 25 percent deflection.

4.4.2.1 Specimen. The specimen shall be a radially cut section of the grommet of 1 inch inside arc length.

4.4.2.2 Procedure. Method 12151 of FED-STD-601 shall be used. The force in pounds required to compress the specimen 25 percent of the cross-sectional diameter shall be reported.

4.4.3 Buoyancy. The plastic grommet shall be suspended in a horizontal position approximately 6 inches beneath the surface of the water by attaching a suitable sinker. After 30 minutes immersion, the weight of the grommet plus the sinker under water shall be determined. The weight of the sinker under water shall also be determined. The difference between these two weight determinations is the net buoyancy of the grommet.

4.4.4 Weight. The weight of the plastic grommet shall be determined on a balance capable of weighing to the nearest 0.1 gram.

4.4.5 Strength in tension, initial. Two pieces of rope 1/4 inch thick shall be attached to the grommet 180° apart, each with two turns around the grommet. A 200 pound weight shall be attached to the free end of one rope. The grommet and weight shall then be gradually lifted by the other rope until the weight is hanging free. The assembly of grommet and weight shall be held in this position for 1 minute, whereupon the grommet is detached. Breakage, tears or cracks, of the grommet shall constitute failure to meet the initial strength requirement.

4.4.6 Strength in tension after 1 week at 165°F. The grommet shall be heat aged in a circulating air oven at 165° + 2°F (73.9° + 1.1°C) for 1 week. The grommet shall then be allowed to stand for 5 hours at 70° + 2°F (21.1° + 1.1°C), and again be tested as specified in 4.4.5, except that a 150 pound weight shall be used. Failure shall be as defined in 4.4.5.

4.4.7 Volume loss on heat aging. The volume loss of a plastic grommet before and after the heat aging test shall be determined by measuring the volume of water displaced by the grommet. The temperature of the water shall be the same for the two measurements. The grommet shall be laid flat on an 8 by 8 inch or larger piece of coated cloth, conforming to type 5 of MIL-C-17415, in an oven maintained at 165° + 2°F (23.9° + 1.1°C) for one week. At the end of this period the coated cloth with the grommet lying thereon shall be removed from the oven and permitted to stand for 5 hours at 70° + 2°F (21.1° + 1.1°C). The grommet shall then be carefully lifted from the cloth. Any adhesion between the grommet and the cloth shall be noted (see 4.4.8). The volume of the grommet shall then be determined as described above.

4.4.8 Blocking after one week at 165°F. Adhesion of the grommet to the coated cloth in the test of 4.4.7 shall constitute failure of the grommet to meet the requirement for blocking.

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4.4.9 Oil resistance.

4.4.9.1 Specimens. Two identical specimens of convenient size each with an arc length of not less than 1 inch shall be cut from a grommet.

4.4.9.2 Procedure. One specimen shall be immersed in ASTM No. 3 oil of ASTM D-471 for 70 hours. The specimen shall then be removed, dipped in alcohol and blotted with filter paper. The treated and untreated specimen shall be compared for apparent softeness, visible swelling and deterioration.

4.4.10 Fire resistance.

4.4.10.1 Specimens. The test specimen shall be 1/4 inch in thickness, 1/2 inch in width and approximately 6 inches in length. Specimens shall be cut from the grommet.

4.4.10.2 Procedure. The specimens shall be clamped at one end in a position such that the long dimension forms a 45-degree angle with the horizontal and with the 1/2 inch width in a vertical position. A candle flame shall be applied to the lower or free end of the specimen for 15 seconds. The candle shall then be removed and the time that the specimen continues to burn after removal of candle shall be recorded as flame duration time. The length of char shall also be recorded. The candle shall conform to class B of C-C-91, and the test shall be performed in a location free from drafts.

4.5 Examination of preparation for delivery. The packaging, packing and marking shall be examined in accordance with Table IV for compliance with the requirements of Section 5. The sample unit shall be one shipping container, fully packed, taken at random just prior to the closing operation. One shipping container fully prepared for shipping shall be examined for closure defects. Sampling shall be in accordance with MIL-P-116.

Table IV - Examination of preparation for delivery.

Examine	Defect
Packaging	Not providing adequate protection during shipment and storage for early use; not level specified; not in accordance with method III of MIL-P-116.
Packing	Not assuring acceptance by common carrier at lowest applicable rate; not affording adequate protection during direct shipment from supply source to using activity not conforming to Uniform Freight Classification Rules or other applicable carrier regulations; not conforming to specified level; container not as specified; case liners missing (when applicable) nor not as specified; not closed and sealed as specified; corners, edge seams and manufacturer's joints not sealed as specified (when applicable); boxes not closed, strapped or banded as specified.
Weight of boxes	Exceeding weight limitation of applicable box specification.
Count per shipping container	Number of pieces per shipping container not as specified.
Marking	Shipment marking not provided on packages and/or shipping containers, incomplete, illegible, incorrect; not in accordance with MIL-STD-129.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early material use.

5.1.1 Packaging. Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment from the supply source to the using activity and until early use and may conform to the supplier's commercial practice when such meets these requirements.

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5.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early use. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements. The number of pieces per container shall be specified, if required (6.2).

5.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include number of grommets, nomenclature, Federal stock number or manufacturer's part number, lot number, date (month and year) of manufacture, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.2).

5.2.1.1 Packaging.

5.2.1.1.1 Level A. Each grommet shall be packaged in accordance with method III of MIL-P-116.

5.2.1.1.2 Level C. Packaging shall be sufficient to afford adequate protection against deterioration and physical damage during shipment, from the supply source to the first receiving activity for early installation and may conform to the supplier's commercial practice.

5.2.1.2 Packing.

5.2.1.2.1 Level A. Grommets shall be packed in containers conforming to any one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-636	Class - weather resistant
PPP-B-640	Class 2

Shipping containers shall have case liners conforming to MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Caseliners for fiberboard boxes may be omitted provided all corners and edge seams and the manufacturer's joint are sealed with minimum 2 inch wide tape conforming to PPP-T-60 or PPP-T-76. Boxes shall be closed and reinforced in accordance with the applicable box specification or appendix thereto except reinforcing shall be by pressure sensitive, reinforced filament tape. The gross weight of shipping containers shall not exceed the weight limitation of the applicable box specification.

5.2 1.2.2 Level B. Grommets shall be packed in boxes conforming to any of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-636	Class - as specified
PPP-B-640	Class 1

Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of shipping containers shall not exceed the weight limitations of the applicable box specification.

5.2.1.3 Marking. In addition to any special marking required by the contract or order, interior and exterior containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Grommets are intended as components of inflatable boats. They are used to carry a life line attached to the grommet which is thrown to personnel in the water for rescue purposes.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Quantity required.
(c) Packaging, packing and marking requirements other than those required by 5.2 (5.3).

6.3 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS,ETC.) PRECLUDES THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 5325-0190)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES. ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

(NAVSHIPS 0VPRT 19-66)

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